

```
In [1]: import pandas as pd
```

```
In [2]: pd.__version__
```

```
Out[2]: '2.3.3'
```

```
In [3]: store = pd.read_csv(r'/Users/abhiramireddygalib/Downloads/Sample - S
```

```
In [4]: store
```

Out[4]:

	Category	City	Country/Region	Customer Name	Manufacturer
0	Office Supplies	Houston	United States	Darren Powers	Message Book
1	Office Supplies	Naperville	United States	Phillina Ober	GBC
2	Office Supplies	Naperville	United States	Phillina Ober	Avery
3	Office Supplies	Naperville	United States	Phillina Ober	SAFCO
4	Office Supplies	Philadelphia	United States	Mick Brown	Avery
...	...	...	...	...	...
10189	Office Supplies	New York City	United States	Patrick O'Donnell	Wilson Jones
10190	Office Supplies	Fairfield	United States	Erica Bern	GBC
10191	Office Supplies	Loveland	United States	Jill Matthias	Other
10192	Technology	New York City	United States	Patrick O'Donnell	Other
10193	Office Supplies	Charlottetown	Canada	Harry Olson	Wilson Jones

10194 rows x 19 columns

```
In [5]: id(store)
```

```
Out[5]: 5832716512
```

```
In [6]: len(store)
```

```
Out[6]: 10194
```

```
In [7]: store.shape # it will give rows and columns
```

```
Out[7]: (10194, 19)
```

```
In [8]: store.columns
```

```
Out[8]: Index(['Category', 'City', 'Country/Region', 'Customer Name', 'Manufacturer',  
              'Order Date', 'Order ID', 'Postal Code', 'Product Name', 'Region',  
              'Segment', 'Ship Date', 'Ship Mode', 'State/Province', 'Sub-Category',  
              'Discount', 'Profit', 'Quantity', 'Sales'],  
             dtype='object')
```

```
In [9]: len(store.columns)
```

```
Out[9]: 19
```

```
In [10]: store.Category
```

```
Out[10]: 0      Office Supplies  
         1      Office Supplies  
         2      Office Supplies  
         3      Office Supplies  
         4      Office Supplies  
         ...  
        10189 Office Supplies  
        10190 Office Supplies  
        10191 Office Supplies  
        10192      Technology  
        10193 Office Supplies  
        Name: Category, Length: 10194, dtype: object
```

```
In [11]: store.isnull()
```

Out[11]:

	Category	City	Country/Region	Customer Name	Manufacturer	Order Date	Or
0	False	False	False	False	False	False	Fi
1	False	False	False	False	False	False	Fi
2	False	False	False	False	False	False	Fi
3	False	False	False	False	False	False	Fi
4	False	False	False	False	False	False	Fi
...	...	...	...	...	...	...	
10189	False	False	False	False	False	False	Fi
10190	False	False	False	False	False	False	Fi
10191	False	False	False	False	False	False	Fi
10192	False	False	False	False	False	False	Fi
10193	False	False	False	False	False	False	Fi

10194 rows x 19 columns

In [12]: store.isnull().sum()

```
Out[12]: Category      0
City      0
Country/Region  0
Customer Name  0
Manufacturer  0
Order Date  0
Order ID    0
Postal Code  0
Product Name  0
Region      0
Segment     0
Ship Date   0
Ship Mode   0
State/Province  0
Sub-Category  0
Discount    0
Profit      0
Quantity    0
Sales       0
dtype: int64
```

In [13]: store[:]

Out[13]:

	Category	City	Country/Region	Customer Name	Manufacturer
0	Office	Houston	United States	Darren	Message

	Supplies			Powers	Book
1	Office Supplies	Naperville	United States	Phillina Ober	GBC
2	Office Supplies	Naperville	United States	Phillina Ober	Avery
3	Office Supplies	Naperville	United States	Phillina Ober	SAFCO
4	Office Supplies	Philadelphia	United States	Mick Brown	Avery
...	...	...	...	...	...
10189	Office Supplies	New York City	United States	Patrick O'Donnell	Wilson Jones
10190	Office Supplies	Fairfield	United States	Erica Bern	GBC
10191	Office Supplies	Loveland	United States	Jill Matthias	Other
10192	Technology	New York City	United States	Patrick O'Donnell	Other
10193	Office Supplies	Charlottetown	Canada	Harry Olson	Wilson Jones
10194 rows × 19 columns					

In [14]: store[0:10]

Out[14]:

Category	City	Country/Region	Customer Name	Manufacturer	Order Date
----------	------	----------------	---------------	--------------	------------

0	Office Supplies	Houston	United States	Darren Powers	Message Book	03-01-2020
1	Office Supplies	Naperville	United States	Phillina Ober	GBC	04-01-2020
2	Office Supplies	Naperville	United States	Phillina Ober	Avery	04-01-2020
3	Office Supplies	Naperville	United States	Phillina Ober	SAFCO	04-01-2020
4	Office Supplies	Philadelphia	United States	Mick Brown	Avery	05-01-2020
5	Furniture	Henderson	United States	Maria Etezadi	Global	06-01-2020
6	Office Supplies	Henderson	United States	Maria Etezadi	Rogers	06-01-2020
7	Office Supplies	Athens	United States	Jack O'Briant	Dixon	06-01-2020
8	Office Supplies	Henderson	United States	Maria Etezadi	Ibico	06-01-2020
9	Office Supplies	Henderson	United States	Maria Etezadi	Alliance	06-01-2020

```
In [15]: store[0:20:5]
```

```
Out[15]:
```

	Category	City	Country/Region	Customer Name	Manufacturer	Order Date
0	Office Supplies	Houston	United States	Darren Powers	Message Book	03-01-2020
5	Furniture	Henderson	United States	Maria Etezadi	Global	06-01-2020
10	Office Supplies	Henderson	United States	Maria Etezadi	Southworth	06-01-2020
15	Office Supplies	Huntsville	United States	Vivek Sundaresam	Acco	07-01-2020

```
In [16]: store.head()
```

Out[16]:

	Category	City	Country/Region	Customer Name	Manufacturer	Order Date
0	Office Supplies	Houston	United States	Darren Powers	Message Book	03-01-2020
1	Office Supplies	Naperville	United States	Phillina Ober	GBC	04-01-2020
2	Office Supplies	Naperville	United States	Phillina Ober	Avery	04-01-2020
3	Office Supplies	Naperville	United States	Phillina Ober	SAFCO	04-01-2020
4	Office Supplies	Philadelphia	United States	Mick Brown	Avery	05-01-2020

In [17]: store.head(3)

Out[17]:

	Category	City	Country/Region	Customer Name	Manufacturer	Order Date	C
0	Office Supplies	Houston	United States	Darren Powers	Message Book	03-01-2020	2
1	Office Supplies	Naperville	United States	Phillina Ober	GBC	04-01-2020	2
2	Office Supplies	Naperville	United States	Phillina Ober	Avery	04-01-2020	2

In [18]: store.tail()

Out [18]:

	Category	City	Country/Region	Customer Name	Manufacturer
10189	Office Supplies	New York City	United States	Patrick O'Donnell	Wilson Jones
10190	Office Supplies	Fairfield	United States	Erica Bern	GBC
10191	Office Supplies	Loveland	United States	Jill Matthias	Other
10192	Technology	New York City	United States	Patrick O'Donnell	Other
10193	Office Supplies	Charlottetown	Canada	Harry Olson	Wilson Jones

In [19]: `store.tail(10)`

Out [19]:

	Category	City	Country/Region	Customer Name	Manufacturer
10184	Furniture	Quebec City	Canada	Bruce Galang	Other
10185	Office Supplies	Quebec City	Canada	Bruce Galang	Acco
10186	Furniture	New York City	United States	Patrick O'Donnell	Other
10187	Office Supplies	Columbus	United States	Chuck Clark	Eureka
10188	Office Supplies	Fairfield	United States	Erica Bern	Cardinal
10189	Office Supplies	New York City	United States	Patrick O'Donnell	Wilson Jones
10190	Office Supplies	Fairfield	United States	Erica Bern	GBC
10191	Office Supplies	Loveland	United States	Jill Matthias	Other
10192	Technology	New York City	United States	Patrick O'Donnell	Other
10193	Office Supplies	Charlottetown	Canada	Harry Olson	Wilson Jones

In [20]: `store.isna() #isnull and isna is same`

Out[20]:

	Category	City	Country/Region	Customer Name	Manufacturer	Order Date	Or
0	False	False	False	False	False	False	F
1	False	False	False	False	False	False	F
2	False	False	False	False	False	False	F
3	False	False	False	False	False	False	F
4	False	False	False	False	False	False	F
...	...	...	...	...	...	...	...
10189	False	False	False	False	False	False	F
10190	False	False	False	False	False	False	F
10191	False	False	False	False	False	False	F
10192	False	False	False	False	False	False	F
10193	False	False	False	False	False	False	F

10194 rows x 19 columns

Statistical concept on pandas

In [21]: `store.describe()` # it gives numerical data

Out[21]:

	Discount	Profit	Quantity	Sales
count	10194.000000	10194.000000	10194.000000	10194.000000
mean	0.155385	28.673417	3.791838	228.225854
std	0.206249	232.465115	2.228317	619.906839
min	0.000000	-6599.978000	1.000000	0.444000
25%	0.000000	1.760800	2.000000	17.220000
50%	0.200000	8.690000	3.000000	53.910000
75%	0.200000	29.297925	5.000000	209.500000
max	0.800000	8399.976000	14.000000	22638.480000

In [22]: `store.columns`

```
Out[22]: Index(['Category', 'City', 'Country/Region', 'Customer Name', 'Manufacturer',
              'Order Date', 'Order ID', 'Postal Code', 'Product Name', 'Region',
              'Segment', 'Ship Date', 'Ship Mode', 'State/Province', 'Sub-Category',
              'Discount', 'Profit', 'Quantity', 'Sales'],
              dtype='object')
```

```
In [23]: store['Category']
```

```
Out[23]: 0      Office Supplies
          1      Office Supplies
          2      Office Supplies
          3      Office Supplies
          4      Office Supplies
          ...
10189    Office Supplies
10190    Office Supplies
10191    Office Supplies
10192      Technology
10193    Office Supplies
Name: Category, Length: 10194, dtype: object
```

```
In [26]: store[['Category', 'City', 'Customer Name']]
```

```
Out[26]:
```

	Category	City	Customer Name
0	Office Supplies	Houston	Darren Powers
1	Office Supplies	Naperville	Phillina Ober
2	Office Supplies	Naperville	Phillina Ober
3	Office Supplies	Naperville	Phillina Ober
4	Office Supplies	Philadelphia	Mick Brown
...	...	...	...
10189	Office Supplies	New York City	Patrick O'Donnell
10190	Office Supplies	Fairfield	Erica Bern
10191	Office Supplies	Loveland	Jill Matthias
10192	Technology	New York City	Patrick O'Donnell
10193	Office Supplies	Charlottetown	Harry Olson

10194 rows x 3 columns

Divide the dataset into categorical & numerical

```
In [27]: store.columns
```

```
Out[27]: Index(['Category', 'City', 'Country/Region', 'Customer Name', 'Manufacturer',  
              'Order Date', 'Order ID', 'Postal Code', 'Product Name', 'Region',  
              'Segment', 'Ship Date', 'Ship Mode', 'State/Province', 'Sub-Category',  
              'Discount', 'Profit', 'Quantity', 'Sales'],  
             dtype='object')
```

```
In [28]: store_cat = store[['Category', 'City', 'Country/Region', 'Customer Name',  
                          'Order Date', 'Order ID', 'Postal Code', 'Product Name', 'Region',  
                          'Segment', 'Ship Date', 'Ship Mode', 'State/Province', 'Sub-Category',  
                          'Discount', 'Profit', 'Quantity', 'Sales']]
```

```
In [29]: store_cat
```

Out[29]:

	Category	City	Country/Region	Customer Name	Manufacturer
0	Office Supplies	Houston	United States	Darren Powers	Message Book
1	Office Supplies	Naperville	United States	Phillina Ober	GBC
2	Office Supplies	Naperville	United States	Phillina Ober	Avery
3	Office Supplies	Naperville	United States	Phillina Ober	SAFCO
4	Office Supplies	Philadelphia	United States	Mick Brown	Avery
...	...	...	...	...	...
10189	Office Supplies	New York City	United States	Patrick O'Donnell	Wilson Jones
10190	Office Supplies	Fairfield	United States	Erica Bern	GBC
10191	Office Supplies	Loveland	United States	Jill Matthias	Other
10192	Technology	New York City	United States	Patrick O'Donnell	Other
10193	Office Supplies	Charlottetown	Canada	Harry Olson	Wilson Jones

10194 rows x 15 columns

```
In [31]: store_cat.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Category              10194 non-null  object
1   City                  10194 non-null  object
2   Country/Region        10194 non-null  object
3   Customer Name         10194 non-null  object
4   Manufacturer           10194 non-null  object
5   Order Date            10194 non-null  object
6   Order ID              10194 non-null  object
7   Postal Code           10194 non-null  object
8   Product Name          10194 non-null  object
9   Region                10194 non-null  object
10  Segment               10194 non-null  object
11  Ship Date             10194 non-null  object
12  Ship Mode             10194 non-null  object
13  State/Province        10194 non-null  object
14  Sub-Category          10194 non-null  object
dtypes: object(15)
memory usage: 1.2+ MB
```

```
In [32]: store_cat.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Category              10194 non-null  object
1   City                  10194 non-null  object
2   Country/Region        10194 non-null  object
3   Customer Name         10194 non-null  object
4   Manufacturer           10194 non-null  object
5   Order Date            10194 non-null  object
6   Order ID              10194 non-null  object
7   Postal Code           10194 non-null  object
8   Product Name          10194 non-null  object
9   Region                10194 non-null  object
10  Segment               10194 non-null  object
11  Ship Date             10194 non-null  object
12  Ship Mode             10194 non-null  object
13  State/Province        10194 non-null  object
14  Sub-Category          10194 non-null  object
dtypes: object(15)
memory usage: 1.2+ MB
```

```
In [33]: store.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Category              10194 non-null  object
1   City                  10194 non-null  object
2   Country/Region       10194 non-null  object
3   Customer Name        10194 non-null  object
4   Manufacturer         10194 non-null  object
5   Order Date           10194 non-null  object
6   Order ID             10194 non-null  object
7   Postal Code          10194 non-null  object
8   Product Name         10194 non-null  object
9   Region               10194 non-null  object
10  Segment              10194 non-null  object
11  Ship Date            10194 non-null  object
12  Ship Mode            10194 non-null  object
13  State/Province       10194 non-null  object
14  Sub-Category         10194 non-null  object
15  Discount             10194 non-null  float64
16  Profit               10194 non-null  float64
17  Quantity             10194 non-null  int64
18  Sales                10194 non-null  float64
dtypes: float64(3), int64(1), object(15)
memory usage: 1.5+ MB
```

```
In [34]: len(store.columns)
```

```
Out[34]: 19
```

```
In [35]: len(store_cat.columns)
```

```
Out[35]: 15
```

```
In [37]: store_num = store[['Discount', 'Profit', 'Quantity', 'Sales']]
```

```
In [38]: store_num.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10194 entries, 0 to 10193
Data columns (total 4 columns):
#   Column    Non-Null Count  Dtype
---  -
0   Discount  10194 non-null  float64
1   Profit    10194 non-null  float64
2   Quantity  10194 non-null  int64
3   Sales     10194 non-null  float64
dtypes: float64(3), int64(1)
memory usage: 318.7 KB
```

```
In [39]: print(len(store.columns))
print(len(store_cat.columns))
print(len(store_num.columns))
```


19
15
4

```
In [40]: store_cat.describe()
```

```
Out[40]:
```

	Category	City	Country/Region	Customer Name	Manufacturer	Order Date
count	10194	10194	10194	10194	10194	10194
unique	3	542	2	800	183	1242
top	Office Supplies	New York City	United States	William Brown	Other	05-09-2022
freq	6128	915	9994	41	1940	38

```
In [41]: store_num.describe()
```

```
Out[41]:
```

	Discount	Profit	Quantity	Sales
count	10194.000000	10194.000000	10194.000000	10194.000000
mean	0.155385	28.673417	3.791838	228.225854
std	0.206249	232.465115	2.228317	619.906839
min	0.000000	-6599.978000	1.000000	0.444000
25%	0.000000	1.760800	2.000000	17.220000
50%	0.200000	8.690000	3.000000	53.910000
75%	0.200000	29.297925	5.000000	209.500000
max	0.800000	8399.976000	14.000000	22638.480000

```
In [ ]:
```